



Can intrafirm IT skills benefit interfirm integration and performance?



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ABSTRACT

This study develops a model to examine whether intrafirm, hard and soft IT skills have cross-boundary effects on interfirm collaboration and integration, thus leading to better supply-chain performance. A model with eight hypotheses was developed and tested using data collected from 250 Taiwanese manufacturing firms. Seven hypotheses find empirical support. The results show that intrafirm IT skills indeed can benefit interfirm collaboration and integration and thus, supply-chain performance. Our findings suggest that the value of skilled IT professionals for interfirm integration, supply-chain performance and collaborative relationships between firms is critical for realizing that value.

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1. Introduction

Over the last decade, firms have sought integration with their supply-chain partners to cope with uncertainty and to obtain better supply-chain performance [27,79,98]. Skilled business personnel certainly lie at the heart of interfirm collaboration for the integration. These business personnel from different departments, such as purchasing, play a boundary-spanner role by contributing their skills and knowledge across firms to help interfirm collaboration and integration [42]. Recently, advanced interorganizational systems (IOS) have profoundly changed the nature of interfirm operations in the supply chain [68,69,75]. Firms have recognized the importance of IOS and begun to use IOS to integrate their processes with those of their supply-chain partners to improve their supply-chain operations [68,69,75]. Prior studies also recognize that IOS integration can provide information visibility to mitigate the bullwhip effect [50], reduce the complexity of supply-chain activities [66], and promote flexibility to meet varying business demands [7]. However, it is not simple to integrate supply-chain partners using IOS. It requires the alignment of both processes and technology standards across firms [34] and demands that firms overcome the various difficulties caused by conflicting processes and standards [50].

To resolve these difficulties, skilled personnel from the cooperating firms must work together to address the conflicts.

In the past, business personnel from core departments, such as purchasing, manufacturing, and selling, typically served as the boundary-spanners, and their intense collaboration helped firms to achieve better interfirm integration [88], to improve goal alignment [66], to enable effective information flows, and to streamline logistics [66], thereby improving supply-chain performance. However, in the context of rapidly advancing IT, the role of IT personnel in enabling interfirm integration has become increasingly critical [68,69,75]. In interfirm operations, technology issues inevitably intertwine with business processes. Without skilled IT personnel and their involvement, firms have limited ability to tackle conflicting processes and technology standards [50], likely resulting in low levels of integration and mutual benefits. However, it remains unclear whether firms' internal IT skills can truly have a boundary-spanning effect when supply-chain partners attempt to integrate their information systems and processes. Consequently, whether and how intrafirm IT human capital, and IT skills in particular, can help interfirm collaboration and integration are issues of both academic and practical significance.

In supply chains, fulfilling collaborative objectives requires not only the participation of relevant personnel but also the application of their skills [73]. Although IT personnel tend to be more familiar with structured approaches and IT knowledge [72], recognized as hard skills, firms have increasingly search for business-literate IT personnel who can add value at a strategic level [44]. Although prior studies have recognized the importance

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of the skills of IT personnel [26,28,35,40,54,82,100], there remains a lack of studies examining the impact of IT personnel on interfirm collaboration and integration. Furthermore, many studies have demonstrated a positive—either direct or indirect—effect of interfirm integration on supply-chain performance [48,68,69,75], but such a positive effect may not materialize without initial help from IT personnel's skills in achieving a high level of IT-enabled interfirm integration.

As Youndt et al. [96] and Kirsch [47] argue, hard skills alone are insufficient to address real-world difficulties because soft skills, such as managerial and interpersonal skills, are often needed to address such difficulties. Soft skills allow IT personnel to work more effectively with business personnel and to better translate IT solutions into business solutions. Of course, the hard and soft skills possessed by IT personnel may produce different boundary-spanning effects on overcoming the difficulties that reside in interfirm activities such as IOS integration. By taking an inside-out perspective [71,90] on IT skills, we argue that internal IT skills should have cross-boundary effects when working with other firms' personnel to develop interfirm mechanisms and processes. By focusing on dyadic relationships in the supply chain, this study examines whether a focal firm's (buyers) internal IT (hard and soft) skills can affect its supply-chain performance through improved interfirm collaboration and integration. Overall, this study attempts to answer the following questions: (1) why and how interfirm integration, including IOS integration and process integration, contributes to supply-chain performance; (2) why

and how interfirm collaboration facilitates interfirm integration; and (3) why and how firms' internal IT skills facilitate interfirm collaboration and IOS integration. Table 1 highlights the lack of clear conceptualization and empirical examination of IT skills and their effects, especially in the interfirm setting. The literature is also fragmented by the notion of skills and has largely operationalized skills with proxies such as organization size and cost or with a broader construct such as IT resources. Thus, it remains unclear whether disparate IT skills truly facilitate interfirm collaboration and IOS integration, and whether they do so differently. By distinguishing soft from hard skills, this study provides a better understanding of the importance and contribution of different IT skills in the supply-chain context.

The remainder of this paper is arranged as follows. First, we discuss the theoretical foundations of our research and develop the research model and associated hypotheses. Next, we introduce our methods for collecting and analyzing the data. After discussing our results and their implications, we conclude with the limitations of our research and directions for future research.

2. The research framework

The central goals of supply-chain operations are to address uncertainty and to enhance performance. To achieve those goals, firms must overcome three problems. The first problem involves a lack of visibility and efficient interfirm operations in the supply chain, causing the bullwhip effect and operational inefficiency.

Table 1
Literature on disparate IT skills related to IOS or interfirm collaboration.

Author	Research topic	Research approach	Related argument or implicit claim involved with skills	Investigation of disparate skills
Fawcett et al. [26]	The influence of interfirm collaboration on supply-chain performance	In-depth interview for concept development	- Investments in people's skills and emotional safety are needed to establish a collaborative culture. - Companies need to build the unique skills and structures for continuous collaborative improvement.	The concept of IT skills is not empirically examined.
Folinas et al. [28]	Analyzing dimensions and key elements within stages of supply chain evolution	Concept development	A number of stages in the evolution of the supply chain, ranging from logistics to collaboration, require soft skills, management skills, or hard skills.	The concept of IT skills is not empirically examined.
Gunasekaran & Ngai [35]	The influence of IT integration on SCM performance	Concept development	IT skills with EDI are the building blocks of infrastructure for SCM.	The concept of IT skills is not empirically examined.
Hoegl & Wagner [40]	The influence of supply-chain collaboration on product-development projects	Empirical survey	Social and project-management skills are important for ensuring that supplier involvement yields the desired collaborative benefits.	The influences of skills on collaboration and performance are not examined. The notions of skills are only discussed in research implications.
de Leeuw & Fransoo [54]	The antecedents of interfirm collaboration	Case study	The more that suppliers recognize skills and capabilities, possess proprietary technology, and are very active in research, the more strategic partnerships are desired.	The concept of IT skills is not empirically examined.
Soosay et al. [82]	The influence of interfirm collaboration on innovation performance	Case study	Competencies formed by partners' skills and expertise are the antecedents of interfirm collaboration.	The concept of IT skills is not empirically examined.
Zhu et al. [100]	The influence of network effect and adoption cost on IOS adoption	Empirical survey	- The use of EDI requires special technical skills. - The implementation of EDI requires firms to develop special technical skills. - Experienced firms have developed certain IOS, technical, and managerial skills.	Skills are acknowledged as adoption costs and switching cost without referring specifically to skills.

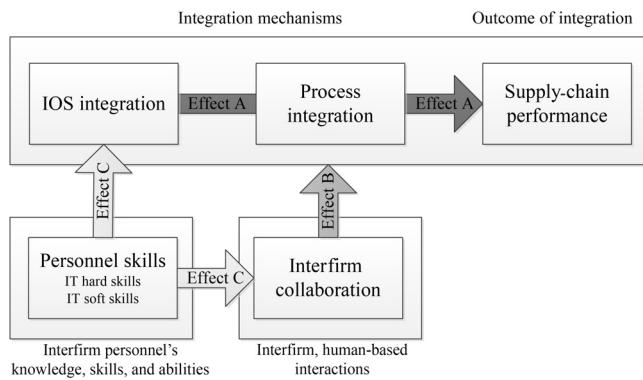


Fig. 1. Research framework.

Firms may cope with such uncertainty through real-time information sharing and smooth interfirm processes, making a highly integrated IOS desirable. The second problem pertains to resolving data conflicts and processing inconsistencies related to the integration. Dealing with these difficulties requires interfirm communication, negotiation, and adaptation. Human-based interactions, such as information sharing and knowledge exchange, therefore are necessary for problem-solving and mutual adaptation. The third problem involves whether firm personnel can effectively perform interfirm communication and negotiation, thus supporting their firms' efforts to address the difficulties of developing interfirm integration mechanisms. Without skilled personnel, firms may underperform in their human-based interactions, thus causing the integration to fail.

Accordingly, a research framework is proposed as depicted in Fig. 1. The framework suggests the following: (1) interfirm operational problems can be mitigated with an integrated IOS, which then enables process integration to produce better supply-chain performance (Effect A in Fig. 1); (2) interfirm human-based interactions can facilitate interfirm integration mechanisms and performance (Effect B); (3) the skills of IT personnel can facilitate interfirm human-based interactions that assist in developing interfirm integration mechanisms (Effect C). The definitions of the research constructs are summarized in Table 2, and the relationships among the building blocks of the framework are addressed from the perspectives of the relational view and human capital theory.

2.1. The relational view and interfirm integration and collaboration

Over the past two decades, interfirm relationships have evolved from arms-length transactions to partnerships to address uncertainty and enhance performance in the supply chain because

partnerships are better at responding to dynamic and unpredictable changes [20]. This evolution from adversarial relationships to “win-win” partnerships is well documented in the academic and trade press [20,24]. The primary motive behind forming a partnership is to reduce uncertainty, thereby gaining costs, cycle time, and quality advantages [50]. To obtain those benefits, partners must integrate human and technical resources without the burden of financial ownership [22]. However, traditional supply chains are a sequence of weakly connected activities and decisions both within and outside of firms, in which a lack of cohesion undermines supply chains' value creation [31]. Thus, supply-chain partners have begun to develop and implement relational mechanisms to address uncertainty, including integrated interfirm processes, real-time information sharing, and intense interfirm collaboration. These mechanisms allow firms to work jointly with greater success than they would find by acting in isolation [94], a phenomenon that the relational view terms relational rents [24]. Dyer and Singh [24] further elaborate on the key sources of relational rents, which include (1) investments in relation-specific assets, (2) joint learning through knowledge-sharing routines, (3) assessing to complementary resources and capabilities, and (4) building up effective governance. Following the logic of the relational view, this study contends that relational mechanisms, including IOS integration, process integration, and interfirm collaboration, are pivotal for supply-chain partners to reduce uncertainty and improve supply-chain performance. Thus, this study focuses on (1) and (2) as the key sources of relational rents.

2.1.1. IOS integration

IOS integration reflects tighter linkages between trading partners' information systems [34]. It requires dyadic firms to mutually develop data consistency and cross-functional applications integration [68,75]. Although data consistency pursues common data definitions and consistency in stored data across partners, applications integration aims at generating cross-functional information visibility to facilitate the real-time coordination of interfirm processes [68]. This integration of data and applications demands resource investments such as IT personnel and often are relation-specific, making IOS integration a relation-specific asset [24]. By improving interfirm information sharing and coordination, IOS integration therefore should be an important relational mechanism for integrating the supply chain.

2.1.2. Process integration

Process integration refers to the management of various sets of activities by seamlessly linking relevant business processes within and across firms and eliminating duplicate or unnecessary parts of the processes to build a better-functioning supply chain [15]. Specifically, process integration consists of process connectivity

Table 2
The definitions of key constructs.

Constructs	Definitions
Supply-chain performance	Supply-chain performance refers to the outcomes realized by dyadic firms that are attributable to their pursuit of improved cost, time, quality, delivery, and flexibility.
Process integration	Process integration refers to the management of various sets of activities by seamlessly linking relevant business processes within and across firms and eliminating duplicate or unnecessary parts of the processes for building a better-functioning supply chain.
IOS integration	IOS integration refers to the tighter linkages between trading partners' information systems.
Interfirm collaboration	Interfirm collaboration refers to supply-chain members sharing information/knowledge and jointly involving themselves in human-based activities to pursue shared strategic goals.
IT hard skills	IT hard skills refer to a set of extant technical skills possessed by IT personnel, such as programming, database design, and network configuration.
IT soft skills	IT soft skills refer to a set of less-structured, loose-normative and non-technical abilities possessed by IT personnel and used to address IT-related work problems or tasks, such as interpersonal skills.

and simplification. Whereas process connectivity refers to smooth linkages between different business processes within and across firms, process simplification is about eliminating unnecessary parts or steps of connected processes [5]. However, achieving process integration is not simple because firms inherently have many heterogeneous processes for their idiosyncratic needs. Firms must tailor their own processes to develop mutually acceptable interfirm processes in which they must invest relevant resources and capabilities collaboratively [24]. Successful process integration enables more effective monitoring and control of process flows and brings about various benefits [49,78]. Process integration thus becomes a valuable relational asset specific to partners [24]. Consequently, integrating interfirm processes toward a seamless supply chain has become a high priority for firms pursuing better supply-chain performance [21].

As relation-specific assets, IOS and process integration enable real-time information sharing and smooth interfirm processes, allowing dyadic firms to work as a single entity, reducing supply shocks, and gaining supply-chain performance (Effect A). Interfirm integration thus embodies the interfirm processes that generate the relational rents, as suggested by the relational view [24].

2.1.3. Interfirm collaboration

Dyer and Singh [24] highlight knowledge-sharing routines, which can install additional capabilities into organizations, as an important dimension of interfirm relationship management [89]. Dyer and Singh [24] argue that a firm's partners are the most important source of new ideas and information that can lead to performance-enhancing technology and innovations. They suggest that superior interfirm knowledge-sharing routines that promote partners' human-based interactions can generate relational rents. Interfirm knowledge-sharing routines may involve strategic thinking, relevant information exchange, human-based activities, and joint problem solving, which collectively imply interfirm collaboration. These routines forge an environment that promotes knowledge sharing and nurtures joint problem solving. Such routines thus assist in addressing the difficulties of integration and creating solutions for better supply-chain performance (Effect B).

Furthermore, interfirm collaboration is a process that requires "a high level of purposeful cooperation" [84,p.77]. This means that members of a supply chain must see the big picture in planning to obtain larger gains from their collaboration [77]. Thus, collaboration between supply-chain members facilitates both operational and strategic foci, allowing them to exploit individual competences and to strengthen the entire supply chain [22]. With supply-chain members as the stakeholders [1], the domain of collaboration is claimed to include information sharing, joint planning, joint problem solving, joint performance measurement, and leveraging resources and skills [59]. Simatupang and Sridharan [77,78] emphasize that interfirm collaboration has three important dimensions: information sharing, decision synchronization, and incentive alignment. These dimensions, involving intense communication and cooperation across all parties in the supply chain, are highlighted as important human dimensions [35]. In particular, interfirm collaboration is characterized by regular formal and informal meetings at which supply-chain members can monitor progress, reassess goals and objectives, discuss collaboration results and action plans, and identify future business opportunities [22]. This explicit and tacit knowledge sharing inevitably is largely facilitated by human interactions. Consequently, we define interfirm collaboration as the condition in which supply-chain members share information and knowledge and jointly involve themselves in human-based activities to pursue shared strategic goals. This definition distinguishes interfirm collaboration from process integration because the latter primarily focuses on process

connectivity and simplification. In addition, the information shared with IOS largely consists of formatted operational data, whereas human interactions allow the sharing of unstructured information and knowledge such as interpretations, opinions, judgments, and thoughts, thus distinguishing interfirm collaboration from IOS integration.

Overall, we adopt the relational view to explain how and why interfirm integration and collaboration as relational mechanisms can generate relational rents, i.e., supply-chain performance [24]. Specifically, we identify interfirm integration as including IOS integration and process integration and interfirm collaboration as embodying knowledge-sharing routines. Our framework therefore suggests that supply-chain performance can be enhanced by process integration, which itself can be enhanced by IOS integration. Although this relationship is consistent with Rai et al. [68] and Saraf et al. [75], we further extend their perspectives by arguing that interfirm collaboration can facilitate interfirm integration, thus highlighting the importance of human-based interaction. Consequently, our research framework reflects the effects of relation-specific assets and knowledge-sharing routines on creating relational rents, which is the primary tenet of the relational view [24].

2.2. Human capital theory and personnel skills

Although interfirm integration inherently involves many difficulties, personnel across firms must address those difficulties by harnessing their skills, for example, managerial or interpersonal skills. Similar difficulties also arise when developing interfirm collaboration. Without the input of knowledgeable and skilled personnel, interfirm integration and collaboration are likely to fail. Accordingly, we draw on human capital theory to complement the relational view and address what individual personnel, particularly IT personnel, can contribute to interfirm integration and collaboration.

Human capital lays the micro foundation for firms to achieve competitive advantage [56]. The central tenet of human capital theory is that personnel who possess knowledge, skills, and abilities provide economic value to firms through increased productivity [96]. Becker [5] further elaborates that human capital can be categorized into general human capital and specific human capital. General human capital, such as work experience, is transferable across different firms, whereas specific human capital, such as firm-specific experience, is productive only at specific firm and is non-transferable across firms. However, the literature also highlights that the mobility of human capital across firm boundaries can facilitate firms' ability to transfer resources or learn [11], resulting in firm performance, growth, and survival [86]. With that mobility, personnel from partner firms can work together to share their firm-specific knowledge, thus creating value specific to the partnership and generating relational rents.

Furthermore, intangible works, such as problem solving, coordination, and judgment, occupy a large portion of firms' daily operations, particularly when they are addressing uncertainties [80]. These works require skilled personnel who can adapt to various technological and strategic changes [14,76]. As Spenner [84] argues, technological and strategic changes can increase productivity but require a broader variety of skills and higher average skills from the workforce. In the supply-chain context, firms tend to utilize relational mechanisms such as collaboration and integration to span organizational boundaries for the purpose of addressing supply-chain uncertainties and enhancing performance. Such relational mechanisms help supply-chain partners synthesize their human capital to co-create value. However, because these mechanisms increasingly depend on information systems, developing them requires skilled personnel—in particular, skilled IT

personnel. Under such circumstances, firms need their IT personnel to possess both technical and problem-solving skills [96] and to use these skills across organizational boundaries. When IT personnel engage in activities related to human-based collaboration and IOS integration, they are expected to possess not only technical hard skills but also soft skills for problem-solving and interpersonal communication. Because supply-chain operations increasingly rely on IT, it is critical for IT personnel to have the ability to work effectively with staff from other companies to resolve problems that arise out of the supply chain [5]. In other words, IT personnel should be able to span their skills across organizational boundaries to enable efficient IT-based, interfirm integration. Accordingly, in an interfirm relationship, IT personnel with hard and soft skills should be transferable human capital that benefits the realization of interfirm integration [5] (Effect C).

Overall, the relational view [24] and human capital theory [96] are complementary in providing an integrated perspective for the current study. Based on the relational view, interfirm integration, including IOS integration and process integration, is the relation-specific asset developed for pursuing better supply-chain performance. Because interfirm collaboration is in the nature of a knowledge-sharing routine, it adds the human effect to the supply chain's value co-creation process. Furthermore, human capital theory [5] provides the basis for the effect of the skills of IT personnel on interfirm collaboration and IOS integration. Although the relational view suggests that interfirm integration and collaboration are important boundary-spanning activities for supply-chain performance [24], human capital theory elaborates that transferable IT skills can be the antecedents of boundary-spanning activities [5,56,70]. Based on these theoretical perspectives, our research model is depicted in Fig. 2. Below, we develop our research hypotheses.

3. Research model and hypotheses

3.1. Process integration and supply-chain performance

Supply-chain performance is a complex concept with many possible dimensions, and it is challenging to select an appropriate measure [27]. Flynn [27] suggests that cost, time, quality, delivery, and flexibility are the important aspects of operational performance. Accordingly, we define supply-chain performance as the outcomes realized by dyadic firms attributable to their pursuit of greater advantages on the aspects of cost, time, quality, delivery, and flexibility. These outcomes may also be classified into tangible and intangible benefits [48]. Tangible benefits include economic outcomes such as lower operating costs, increased productivity, and improved asset management [48]. Intangible benefits are outcomes that are difficult to quantify, such as improved production planning and time to market, improved resource control, and increased flexibility [48].

As the main tenet of the relational view [24], partnership is key for creating mutual benefits between firms. Sanders [74] also

suggests that managing a supply chain as a single entity is critical to superior performance. Accordingly, process integration that treats dyadic firms as a single entity and eliminates operational barriers between them enhances supply-chain performance. By interconnecting and simplifying processes, process integration can improve the efficiency of material and product transportation and reduce mistakes in interfirm operations. It also enables firms to improve productivity, increase order frequency, cut inventory, and reduce purchasing costs. Process integration thus enhances the agility of the supply chain [66] and thus, its responsiveness to the market [68]. Moreover, the goal of process integration is to seamlessly create and coordinate interfirm, relation-specific processes that competitors cannot easily match [97]. Not only is the performance effect of process integration well documented in the literature [53] but also process integration can be a source of long-term competitiveness [68]. Thus, we propose:

H1. Process integration is positively associated with supply-chain performance.

3.2. IOS integration and process integration

According to the resource-based view (RBV) and its later developments, resources must be transformed into higher-order capabilities to ensure both competitive advantage and organizational performance because capabilities are more difficult to imitate than resources [33]. Recent IS research has reframed the notion from the direct performance effects of IT resources to how IT shapes higher-order process capabilities to create firm performance [4,60]. In the supply-chain context, when IOS integration is viewed as a configured IT resource/capability, process integration can be viewed as a higher-order capability enabled by IOS integration [68]. The rationale is that information sharing is the core of interfirm process integration because data exchange underlies material processes [6]. The enabling technology, which provides accurate, timely, and relevant information to parties in the supply chain, can support the management and control of material flows across organizational boundaries [1]. IOS integration therefore tightens the information linkages between firms to support their physical linkages, enabling them to operate either as a single entity or in a partnership relationship [24]. Past studies also suggest that IOS integration not only improves information sharing in the supply chain and minimizes the bullwhip effect [95] but also enables process changes [75]. An integrated IOS can eliminate redundant steps to simplify processes. It can coordinate disparate processes and integrate data to allow many sequential processes to be handled in parallel [7]. Thus, streamlined information flows that support timely information sharing to enable interfirm process integration are the core of IOS integration. Consequently, we propose the following hypothesis:

H2. IOS integration is positively associated with process integration.

3.3. Interfirm collaboration and supply-chain performance

While removing unnecessary steps and speeding up information and material flows are critical operational dimensions of supply-chain performance, a long-term collaborative partnership is the foundation of continuous supply-chain performance improvement [97]. Many studies have indicated that interfirm collaboration offers the promise of improving supply-chain performance in areas such as sales, cost and inventory management, forecasts, and customer service [45,61]. With the knowledge-sharing routines that are embedded in collaboration, partners are encouraged to exchange explicit and tacit knowledge to generate new knowledge for their

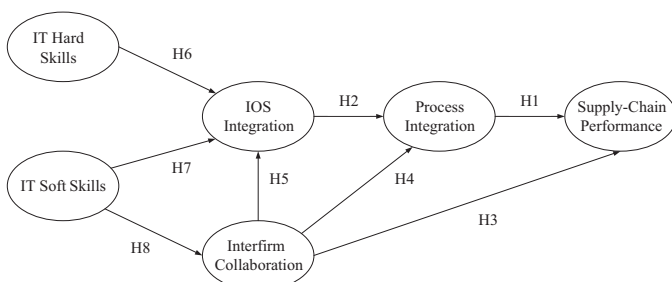


Fig. 2. Research model.

mutual benefit [51]. For example, collaborative effort for better logistics design has the potential to reduce costs and improve responsiveness, and sharing information with suppliers benefits available-to-promise and shortens stock-out waiting time [31]. Experiments also have demonstrated that interfirm collaboration improves supply-chain performance in terms of better stability and service level [31]. Furthermore, interfirm collaboration can reduce the costs of opportunism and monitoring, thus increasing the probability that supply-chain partners will behave in the partnership's best interest [12]. A recent study by American Marketing Association Research indicates that interfirm collaboration can add as many as three percentage points to profit margins for all types of supply-chain players [3]. Consequently, we propose the following:

H3. Interfirm collaboration is positively associated with supply-chain performance.

3.4. Interfirm collaboration and process integration

Integrating processes across firms inevitably requires human collaboration to exchange opinions, negotiate resources, resolve conflicts, and formulate solutions, because investment in relation-specific assets may require intense interfirm knowledge sharing [24]. Process integration may also involve continuous splitting and combining processes, demanding significant collaboration between partners [63]. A collaborative relationship will make it easier to simplify and connect business processes across firm boundaries [15]. Collaboration also generates extensive communication, both formal and informal, and nurtures shared values [30], helping partners to address and resolve the difficulties of process integration. Common wisdom also suggests that partners who share business information collaboratively can integrate their processes more effectively by simplifying core processes, streamlining interfirm operations, and reducing consequent channel-wide costs [16,57]. Thus, we propose the following:

H4. Interfirm collaboration is positively associated with process integration.

3.5. Interfirm collaboration and IOS integration

Systems integration is all about interoperability [61,76]. Implementing IOS integration inevitably must address various technology problems caused by conflicts in networks, data, and applications of the integrating firms [61]. To resolve conflicts or incompatibilities, collaboration between partners is necessary. When partners' IT personnel can address technical issues collaboratively, goal alignment and opinion feedback are more likely to occur [30,66]. Such collaboration promotes the willingness of IT personnel to share tacit and idiosyncratic information and relevant risks and therefore helps in overcoming the difficulties of IOS integration. In addition, when interfirm transactions evolve from simple to complex forms, collaboration that is more intensive naturally is helpful in addressing the increased complexity necessary to achieve a higher level of IOS integration. Thus, we propose the following hypothesis:

H5. Interfirm collaboration is positively associated with IOS integration.

3.6. IT hard skills and IOS integration

Many studies on innovation adoption claim that large organizations are more likely to adopt EDI because they possess the resources and skills necessary to assimilate innovation [87]. For

innovation adoption, availability of skilled personnel and site-specific requirements are noneconomic reasons to emphasize [43]. Hard skills possessed by IT personnel, such as programming, database design, and network configuration, are undoubtedly needed to address the technical issues of network compatibility, data consistency, and applications integration underlying IOS integration. For example, with sufficient programming skills, IT personnel can troubleshoot and fix issues related to applications integration or create necessary middleware. Similarly, database design skills are useful for dealing with the data inconsistency between firms. These IT hard skills as technical expertise thus should be transferable to and useful in the IOS integration context [70]. Consequently, we propose the following:

H6. IT hard skills are positively associated with IOS integration.

3.7. IT soft skills and IOS integration

Prior IOS studies largely have presumed that firms possess the necessary skills and focus on non-human factors to account for the effectiveness of IOS integration [62]. However, implementing IOS integration has inherent complexity, and a lack of managerial capability for IOS implementation will increase the risk of more changes, incur greater costs, or obtain fewer benefits than expected [100]. Managerial capabilities/skills are soft abilities and in practice, the appeal to soft skills has long been pervasive. IT soft skills are less-structured, loose-normative and non-technical abilities used to address IT-related work problems or tasks. To address IT soft skills in the supply-chain context, we focus on technology management and interpersonal skills [52] because interfirm collaboration and integration often require technological solutions with human participation. Although technology management skills emphasize how to deploy IT effectively to meet business objectives, interpersonal skills focus on how to achieve success through interactions among individuals. Although these skills have characteristics of general human capital [5], they can nevertheless be applied to interfirm negotiation and problem solving. For IT personnel to address the problems of IOS integration, they need both technology management skills and interpersonal skills because the involvement of partners in management and communication accounts for the success/failure of IOS integration. When IT personnel possess technology management skills such as IS functions management, project management, and understanding of technological trends, they should be able to integrate data and applications, to allocate resources, and to avoid technological risks more effectively. These skills help IT personnel to identify, to negotiate, and to resolve the technical problems of IOS integration with their partner-firm counterparts. Furthermore, interpersonal skills also help IT personnel from different firms to communicate and establish trust with each other when dealing with technical issues. We thus propose the following hypothesis:

H7. IT soft skills are positively associated with IOS integration.

3.8. IT soft skills and interfirm collaboration

Fulfilling collaborative objectives requires not only the participation of the relevant stakeholders but also the application of various skills in the supply-chain environment [73]. Competences and resources for building and maintaining effective relationships with suppliers and customers are the important preconditions of successful collaboration in supply chains [32]. In other words, successful interfirm collaboration is leveraged by joint capabilities and resources across the supply chain [22].

Adequate skills can promote effective negotiations for collaborative activities. Although interfirm collaboration involves explicit and tacit knowledge transfer that resides in “social interactions” [51], boundary-spanning personnel and activities are critical for achieving effective collaboration. This implies the importance of the human aspect of collaboration. Ellinger [25] proposes that the success of collaboration depends on individuals’ ability to build meaningful relationships. Similarly, Braithwaite [9] argues that the development of skills and talents related to the supply chain requires soft interpersonal skills because the concept of interfirm collaboration has as its basic principles relationships and working together. Thus, supply-chain partners must possess sufficient social and project-management skills to ensure mutual involvement and desired benefits [40].

Specifically, IT personnel are also expected to have both business and interpersonal skills [52]. In the supply-chain context, IT personnel are expected to possess soft skills that facilitate interfirm collaboration in addressing IT-related problems. Soft skills allow IT personnel to grasp precise business requirements and nurture fine-grained communications with collaborating partners. Specifically, technology management skills help IT personnel to understand key success factors and follow technology trends, thus ensuring the strategic quality of joint actions/activities. In addition, interpersonal skills, which are required to render cross-functional or project teams effective, have the potential to nurture shared understanding between partners and thus, to promote their willingness to collaborate. Consequently, we propose the following hypothesis:

H8. IT soft skills are positively associated with interfirm collaboration.

4. Research methodology

4.1. Survey administration

A cross-sectional mail survey was administered to collect data from the manufacturing firms in Taiwan. A draft survey was developed primarily based on measures identified in the literature as suitable for this study. After compiling an English-language version of the questionnaire, the original questionnaire was translated into Chinese. The survey items then were verified and refined for translation accuracy by an MIS professor and a senior doctoral student. Two thousand survey packages were mailed on April 18, 2012, to the senior IT managers of the top 2000 manufacturing firms in the directory *Top 5000 Corporations in Taiwan 2010*, published by the China Credit Information Service. IT managers were chosen as respondents because senior IT managers should be the most knowledgeable and reliable informants in a company about the domains of IOS and IT department operations. We also believe that senior IT managers should be familiar with interfirm operations because implementing and maintaining IOS requires an understanding of business transactions and interfirm interactions. This implies that senior IT managers are knowledgeable enough to answer survey questions about interfirm collaboration, process integration, and supply-chain performance. Three hundred surveys were returned and 250 of those were completed and used in our subsequent analysis, yielding an effective response rate of 12.5%. The characteristics of the responding firms are depicted in Table 3. Among the responding firms, more than 70 percent earned less than NT\$ 8 billion revenue in 2011, which reveals that the sample mainly consists of medium-sized firms. On average, the informants had been in their current positions for 6.4 years and in their firms for 10.5 years, indicating that they should have had sufficient knowledge to answer the survey.

Table 3

Demographic characteristics of the responding firms ($n = 250$).

Sample statistics	Percentage of firms
Industry	
Automobile	8.8
Chemical	6.0
Computer and electronics	38.8
Food	2.8
Machine and tool	14.0
Metal and steel	9.2
Textile	7.2
Others	13.2
Previous-year revenue (2011) (NT\$)	
Less than \$1 billion	31.2
\$1–2 billion	26.4
\$2–4 billion	19.6
\$4–8 billion	10.8
\$8–16 billion	6.0
\$16–32 billion	3.2
\$32–64 billion	1.6
More than \$64 billion	0.8
Missing value	0.4
Number of employee	
Fewer than 100	12.4
101–500	62.8
501–1000	11.6
1001–3000	10.0
More than 3000	3.2
Number of IT department employees	
Fewer than 10	66.4
11–30	17.6
31–50	2.4
51–100	2.0
More than 101	10.8
Missing value	0.8

Non-response bias was assessed using the procedure recommended by Armstrong and Overton [2]. Considering the last group of respondents as most likely to be similar to non-respondents, a comparison of the first and last quartile of respondents provided a test of response bias. No significant differences between the first and last quartile of all respondents were found on employee numbers based on χ^2 test ($p = 0.402$). Accordingly, non-response bias may not be a serious concern in this study.

4.2. Measures

IT hard skills were measured using five items adapted from Byrd and Turner [10] and Lee et al. [52], assessing abilities related to programming, systems analysis and design, network, and data storage. IT soft skills focus on IT personnel’s abilities to manage IS functions, interact with users, and perform project management and leadership. We modeled the measure as a second-order formative construct with technology management skills and interpersonal skills as the two dimensions. Technology management skills were measured using five items, assessing how well IT personnel manage IS functions and engage in self-learning related to IT. Eight items were used to measure IT interpersonal skills in socialization, collective work, and cooperation. IOS integration assesses the IS applications of a focal firm that work as a functional whole in conjunction with the IS applications of its major supplier, measured using six items adapted from Saraf et al. [75] and Rai et al. [69]. Interfirm collaboration focuses on interfirm decision-making processes that involve information and knowledge sharing and joint problem solving. We adapted those measures from Stank et al. [85] and Kim et al. [46]. Process integration assesses the extent to which interfirm processes are operationally integrated

Table 4Interconstruct correlations and reliability measures for first-order constructs ($N=250$).

Construct	Items	Mean (Std.)	CR	AVE	Correlations among constructs						
					1	2	3	4	5	6	7
Interpersonal skills (IS)	8	5.66 (0.73)	0.96	0.62	0.79						
Technology management skills (TMS)	5	5.69 (0.73)	0.87	0.58	0.69	0.76					
IT hard skills (HS)	5	5.00 (0.85)	0.85	0.53	0.50	0.51	0.73				
Interfirm collaboration (CO)	6	4.54 (1.18)	0.94	0.73	0.10	0.17	0.47	0.85			
IOS integration (IOSI)	6	2.92 (1.46)	0.96	0.79	0.36	0.45	0.32	0.56	0.89		
Process integration (PI)	5	3.57 (1.54)	0.97	0.86	0.17	0.18	0.28	0.53	0.63	0.93	
Supply-chain performance (SCP)	11	4.93 (1.06)	0.97	0.72	0.30	0.37	0.29	0.36	0.36	0.43	0.85

across firm boundaries. Five items were adapted from Saraf et al. [75]. Supply-chain performance focuses on the aspects of improved asset management, increased productivity, lower operating costs, improved production planning, improved resource control, and increased flexibility. We measured it using eleven items adapted from Klein et al. [48] and Wang et al. [91]. All of the first-order constructs were measured using reflective indicators on a seven-point Likert scale anchored from strongly disagree (1) to strongly agree (7). The items are provided in Appendix A.

4.3. Results

A partial least squares (PLS) structural-equation model was constructed for measurement validation and hypotheses testing because PLS places minimal restrictions on measurement scales, sample size, and residual distribution [17]. We used SmartPLS 2.0 M3 developed by Ringle, C.M., Wende, S. and Will, A in Hamburg, Germany to estimate the parameters in the outer model and the inner model using a path-weighting scheme [36,39]. Following the guidelines in the literature [64,93], the second-order construct of IT soft skills was set up through the repeated use of the items of its first-order constructs.

4.3.1. Measurement model

The adequacy of the measurement model was evaluated on the principles of reliability, convergent validity, and discriminant validity. Reliability was examined by using the indicator loadings and the composite reliability (CR) values. Indicator loadings for most items are above 0.7 and are all significant at $p < 0.001$ level, indicating individual item reliability [36]. Table 4 shows that the CR estimates are above 0.8 for all constructs, indicating good internal consistency and reliability of our scales [36]. The convergent validity of the scales was assessed by the average variance extracted (AVE) for each construct exceeding the minimum threshold value of 0.50 [29,36,38]. As shown in

Table 4, all of the AVE values are above 0.50, thus satisfying the criteria for convergent validity. The above combined results demonstrate the convergent validity of our measures.

Discriminant validity was assessed by two criteria. First, if the loading of each measurement item on its assigned construct is larger than its loadings on all other constructs, it is considered as having good discriminant validity [18]. Second, the square root of the AVE of a construct should be greater than the correlations between the construct and all other constructs in the model [29]. As shown in Table 4 and Appendix B, both criteria are clearly met, demonstrating sufficient construct validity of the scales.

In addition, variance inflation factors (VIFs) were used to assess the degree of multicollinearity. We conducted a regression analysis with supply-chain performance as the dependent variable and the other six variables as the independent variables represented by their un-standardized latent scores. The VIFs ranged from 1.807 to 2.142, which are below the suggested threshold of 3.3 [23]. Therefore, no significant multicollinearity problem exists with regard to our data.

Furthermore, we assessed the validity of IT soft skills as a second-order formative construct based on [13,64] by (1) assessing multicollinearity among the first-order constructs, and (2) examining the path weights and correlations among the first-order constructs and the second-order construct. Multicollinearity was first assessed using the VIF generated by SPSS when regressing the unstandardized latent scores of interpersonal skills and technology management skills on the unstandardized latent scores of the repeated indicators of IT soft skills. The VIF is 1.867, well below the 3.3 threshold [23], indicating no multicollinearity problem. Second, both path weights between the first-order constructs and the second-order constructs are significant (see Fig. 3). The bivariate correlations between IT soft skills and interpersonal skills and technology management skills are 0.694 and 0.385, respectively ($p < 0.01$). The correlations demonstrate strong absolute relationships.

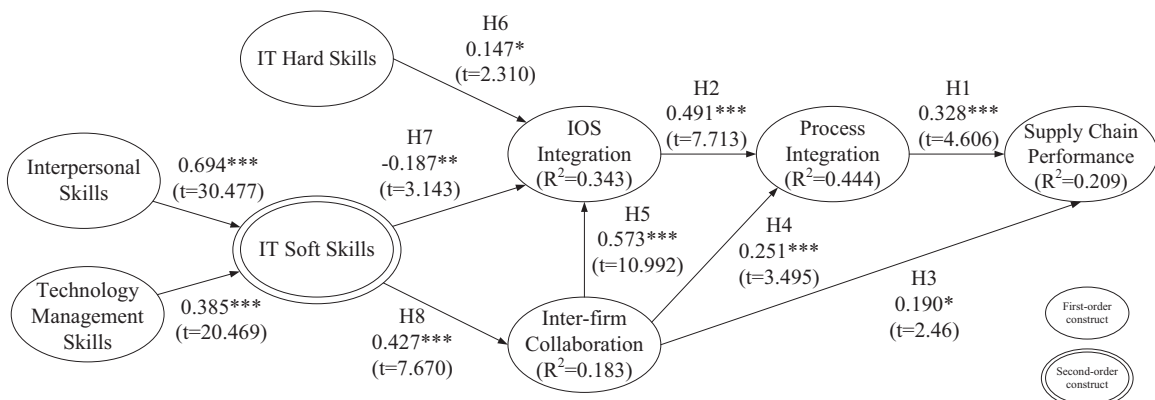
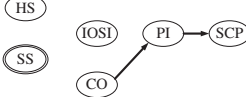
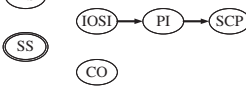
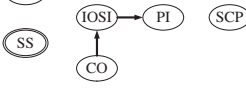
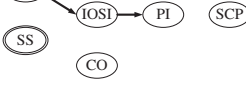
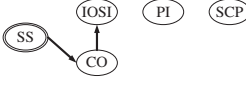
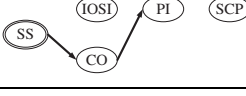


Fig. 3. PLS analysis of results. Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$ (two-tailed test).

Table 5
Significance of mediation paths.

Row	Indirect effect	<i>c</i>	α	β	<i>c'</i>	$\alpha\beta$	Sobel <i>Z</i>	Symmetric 95% CI	Bootstrap 95% CI 99% CI	Type
1		0.32 (.000)	0.68 (.000)	0.23 (.000)	0.17 (.005)	0.15	4.37	0.08, 0.22	0.08, 0.24 0.06, 0.27	Partial
2		0.26 (.000)	0.66 (.000)	0.23 (.000)	0.10 (0.06)	0.15	4.31	0.08, 0.23	0.09, 0.23 0.07, 0.25	Full
3		0.68 (.000)	0.69 (.000)	0.52 (.000)	0.32 (.000)	0.36	6.65	0.25, 0.46	0.24, 0.49 0.21, 0.54	Partial
4		0.51 (.000)	0.52 (.000)	0.63 (.000)	0.18 (.06)	0.33	4.57	0.19, 0.47	0.18, 0.48 0.14, 0.54	Full
5		0.28 (.037)	0.74 (.000)	0.76 (.000)	−0.28 (.028)	0.56	6.04	0.38, 0.74	0.40, 0.73 0.35, 0.79	Partial
6		0.42 (.003)	0.74 (.000)	0.71 (.000)	−0.10 (.438)	0.52	5.72	0.34, 0.70	0.37, 0.70 0.32, 0.77	Full

Note: *c* = the total direct effect of the independent variable on the dependent variable; α = the effect of the independent variable on the mediating variable; β = the effect of the mediating variable on the dependent variable when controlling for the independent variable; *c'* = the effect of the independent variable on the dependent variable when controlling for the mediating variable; $\alpha\beta$ = the total indirect effect on dependent variable; *p* values shown in parenthesis.

Common method variance (CMV) was examined by conducting Harmon's single-factor test [65]. Seven factors with eigenvalues > 1 were extracted; collectively, they accounted for 71.35% of the variances in the data, with the first factor accounting for 33.09% of the variances. These findings suggest that CMV is not a main concern.

4.3.2. Structural model

To assess the explanatory power of the structural model, we examined the structural paths and the R-square scores of endogenous constructs. Fig. 3 shows the results of structural path analysis. All paths exhibited as significant at $p < 0.05$ level. The significance of all paths was assessed using 5000 bootstrapping runs [36]. The R-square scores of the endogenous constructs range from 18.3% to 44.4%, representing good explanatory power in this model. With omission distance equal to 7, all the cross-validated redundancy Q^2 values of endogenous constructs are larger than zero, indicating that the exogenous constructs have predictive relevance for the endogenous constructs under consideration [19]. Thus, the fit of the overall model is fairly good.

4.3.3. Mediation testing

We further used mediation analysis techniques to strengthen our results. We followed the guidelines suggested by Zhao et al. [99] for justifying full or partial mediation and used the mediation regression method with the percentile bootstrap approach to assess the significance of indirect paths [37,99] because that approach is more powerful than Sobel's test [58,81]. We adopted the simple mediation model [67]. Because the approach is regression based, we used the PLS algorithm to obtain

unstandardized latent scores of the research constructs as inputs [8] to perform the mediation regression with a second-order exact solution and 5000 resamplings on SPSS macros provided by Preacher and Hayes [67]. The results are shown in Table 5. As suggested by Zhao et al. [99], we first examined the significance of indirect effect with the simple mediation (single mediator) model. The result indicates that all mediation effects are significant at $p < 0.01$ level. To justify full or partial mediation, we then examined the significance of the direct effects when the mediator was controlled. The result indicates that the mediated paths of 2, 4, and 6 are fully mediated. Overall, our mediation test again confirms the results of the structural model.

5. Discussion and implications

According to the results, seven out of eight hypotheses are supported, except for the effect of IT soft skills on IOS integration, which surprisingly, is negatively significant. As our mediation test shows, the direct effect of IT soft skills on IOS integration is positively significant ($\beta = 0.28$; $p < 0.05$; see Row 5 in Table 5) when excluding interfirm collaboration. However, that direct effect becomes negatively significant ($\beta = -0.28$; $p < 0.05$) when the effect of interfirm collaboration is controlled for. The indirect effect is significant at $p < 0.01$ level (Sobel $Z = 6.0401$; for bootstrap results, zero is excluded in the 99% confidence interval). Following the guideline suggested by Zhao et al. [99], we conclude that the indirect path we proposed could be a "competitive mediation" or that an omitted mediator might be in the direct path. Thus, we argue that trust and bargaining power may play a latent role in confounding the proposed effect. IT cultural values [55] may also

provide an explanation for this surprising result. That is, the broader the range of IT soft skills, such as technology management skills and interpersonal skills, the greater the possibility that IT personnel tend to employ those non-technical skills to avoid the technical problems that are often perceived as hard work according to the values of IT personnel. Such an inclination may motivate IT personnel to avoid such “dirty jobs” as detailed, time-consuming technical solutions in IOS integration. Thus, the willingness to faithfully conduct the required work of IOS integration may decrease, hindering IOS integration. Another possible explanation is that IOS integration often enables sharing of strategy information [48]. Top management may interfere with or even take over an IOS integration project because of the sensitivity of the information involved. This may direct IT personnel's soft skills toward avoiding certain information being shared through IOS, resulting in a less-integrated IOS. Although these may be some tentative reasons for the surprising, negative relationship, further theorizing and empirical investigation are required. Despite this unexpected effect, our results nevertheless suggest that IT soft skills can still positively influence IOS integration indirectly through interfirm collaboration, suggesting the importance of an interfirm collaborative atmosphere for making IT soft skills useful in IOS integration.

As expected, the effect of IT hard skills on IOS integration is significant. This result supports most prior studies suggesting that IT personnel with sufficient technical knowledge can contribute to the implementation of information systems [35,72,100]. In addition to confirming the indirect effects of the skills on process integration, the mediating testing demonstrates that the effects are fully mediated by IOS integration ($c' = 0.18$; $p > 0.05$; see Row 4 in Table 5) and interfirm collaboration ($c' = -0.10$; $p > 0.05$; see Row 6 in Table 5). Thus, the skills influence process integration indirectly, instead of directly, through IOS integration and interfirm collaboration. This implies that process integration may be a higher-order capability that requires other capabilities as its foundations.

Furthermore, our results suggest that supply-chain performance can be enhanced by process integration ($\beta = 0.328$; $p < 0.001$), and process integration can be enhanced by IOS integration ($\beta = 0.491$; $p < 0.001$). These results are consistent with those of Rai et al. [68], Rai and Tang [69], and Saraf et al. [75]. We therefore suggest that the linkage from IOS integration to supply-chain performance through process integration (Sobel $Z = 4.31$; for bootstrap results, zero is excluded in the 99% confidence interval) should be an important interorganizational rent-generating process for dyadic firms [24].

Finally, we find that human-based interfirm collaboration has significant positive effects on supply-chain performance ($\beta = 0.19$; $p < 0.05$), process integration ($\beta = 0.251$; $p < 0.001$), and IOS integration ($\beta = 0.573$; $p < 0.001$). These results provide evidence that interfirm collaboration can benefit supply-chain performance by facilitating interfirm explicit and tacit knowledge sharing [31,51]. The results also imply that intense collaboration is helpful for resolving problems related to process integration and IOS integration. Our results thus contribute to the literature by demonstrating the positive effects of human-based collaboration on interfirm process and IT-based integration.

5.1. Implications for research

Overall, our research framework adopts the relational view to identify interfirm integration—including IOS integration, process integration, and interfirm collaboration—as important enablers of supply-chain performance and applies human capital theory to explore the effect of intrafirm IT skills on those enablers. With these factors reflecting the people, process, and system aspects of

supply-chain partnerships, this study contributes the following insights to the literature.

First, IOS integration enables the process integration of supply chains by allowing supply-chain partners to operate more like single entities, resulting in greater supply-chain performance (Effect A in Fig. 1). With the significant relationships among IOS integration, process integration, and supply-chain performance, the proposed model also suggests that process integration is a key mediator between IOS integration and supply-chain performance. These results support the relational view that investing in relation-specific assets, i.e., IOS integration and process integration, generates relational rents [24]. Although our results are similar to those of prior studies [68,75], Rai et al. [68] take an outside-in perspective and focus on firm performance instead of supply-chain performance. Our study takes an opposite, inside-out perspective of IT skills, enhancing our understanding of the cause and effect of interfirm integration from a different perspective.

Second, based on the relational view, we find that interfirm collaboration with embedded knowledge-sharing routines is the basis for realizing interfirm integration through IOS and processes (Effect B in Fig. 1). Our results demonstrate that interfirm collaboration not only facilitates difficult integration work but also directly contributes to supply-chain performance, confirming the value of the relational view in analyzing supply chains [24]. The direct and indirect effects of interfirm collaboration imply its dual effects at both the operational and strategic levels. This result supports the arguments of Daugherty et al. [22] that collaboration between supply-chain members facilitates both operational and strategic foci.

Third, our results contribute to human capital theory by showing that intrafirm human capital can be utilized and transferred to generate interfirm, relation-specific assets that benefit supply-chain performance. Human capital theory has indicated the significant differences between general and specific human capital [5]. Although IT skills are primarily general human capital and thus transferable to the interfirm context, they have the added benefits of making implicit, firm-specific knowledge explicit, thus facilitating interfirm collaboration and problem solving. That boundary-spanning effect helps the progress of interfirm tasks with supply-chain partners (Effect C in Fig. 1), thus integrating the theoretical perspectives of human capital theory and the relational view. Although most researchers concede that IT skills play an important role in information system implementation and integration in the supply-chain context [35,100], little empirical evidence has been provided. This study thus contributes to the literature by providing theoretical explanations and empirical evidence to fill the knowledge gap.

Finally, we show that different types of IT skills have different effects on boundary-spanning activities. The effects of IT personnel's hard skills, as expected, can facilitate IT-related tasks such as IOS integration. Although most studies argue that soft skills, such as managerial and interpersonal skills, can facilitate problem solving and therefore goal achievement, we find that IT soft skills can affect IOS integration only indirectly, through interfirm collaboration. This finding implies that soft skills can nurture collaboration, which in turn, helps to overcome the resisting forces of organizational inertia and risk aversion to achieve the mutual adjustments required by IOS integration. This suggests that a collaborative atmosphere is pivotal for cooperating firms to span the effect of their intrafirm competences across organizational boundaries to obtain greater relational rents.

5.2. Implication for practice

This study suggests that integrating IOS with process integration can realize greater benefits and thus, practitioners may

prioritize various types of integration to realize greater supply-chain performance. Interfirm collaboration, which contributes to “win-win” strategies, proves its worth for co-creating value between supply-chain partners at both the operational and strategic levels. Without the base of a collaborative relationship, it is not possible to achieve a high level of interfirm integration for greater supply-chain performance. Close collaboration provides opportunities for supply-chain partners to reconfigure their mindsets to enter the “winner zone” in an increasingly competitive business environment.

Our findings also indicate that IT personnel can be important boundary-spanners in facilitating interfirm collaboration and integration. Business personnel, such as purchasing and sales professionals, are often highly ranked by practitioners because of their direct relationships with value creation within and across firms. However, as our research suggests, IT personnel should also be important boundary-spanners, especially for interfirm IT-related tasks. Because today's supply chains are increasingly supported by IT, a lack of skilled IT personnel will limit supply-chain partners' ability to obtain fuller benefits from more effective supply-chain operations. Thus, practitioners should encourage IT personnel to participate in interfirm collaboration and operations and to foster both hard and soft skills. The importance of soft skills has been touted for decades, even for exclusively intrafirm IT development. In the interfirm context, IT personnel with better soft skills are more prepared to contribute to the development of new business processes with appropriate technical solutions that produce more efficient interfirm operations and higher profits. Business practitioners should recognize the important role of IT personnel in seeking greater supply-chain efficiency and effectiveness.

This study further suggests that firms should pay attention to their human resource policies for IT personnel. Firms often ignore the fact that their IT human capital development follows a “deskilling” policy rather than an “upskilling” one [96] due to an overemphasis on technical skills. “Deskilling” refers to the fragmentation of technology use and work to lower the breadth and depth of employees' required skills. In contrast, “upskilling” refers to improving employees' skills, usually through training, so that they will be better at their jobs. The adoption of a human resource policy is generally related to three primary strategies: cost, quality, and flexibility [96]. Cost strategy treats employees as one of the most costly and uncontrollable resources for creating value. Quality strategy considers that employees with superior technical, problem-solving, and interpersonal skills can increase productivity. Flexibility strategy posits that a highly skilled, technologically competent, and adaptable workforce can address non-routine and exceptional circumstances. If a firm adopts a quality or flexibility strategy, it is more likely to adopt an “upskilling” policy for employee development. For IT personnel, we suggest that firms may invest more resources in fostering their IT personnel's soft skills with a quality or flexibility strategy. Soft skills are useful for a relatively longer time than are hard skills because technologies can easily become obsolete given the rapid development of IT.

6. Conclusion

Overall, this study shows that intrafirm IT skills can produce a boundary-spanning effect to benefit interfirm collaboration and integration, enhancing our understanding of the effect of human skills in the interfirm setting. In particular, IT soft skills are distinguished from IT hard skills to enrich our knowledge regarding how distinct human skills may serve differently for better interfirm collaboration and integration and eventually lead to greater supply-chain performance. This research also finds that interfirm collaboration is an effective cross-boundary mechanism in the supply chain due to its simultaneous, beneficial effects on integration and performance. Thus, interfirm collaboration has the potential to provide both operational and strategic benefits in partnerships. We also confirm that process integration is critical in mediating the effect of IOS integration on supply-chain performance.

Our research has a number of limitations, however. First, collecting data from both sides of a dyad demands a great deal of effort and resources. Thus, we chose buying firms in dyadic relationships as respondents because the buying side is closer to customers who stoke the demand for better supply-chain performance. However, this data-collection issue remains a limitation. Second, adopting the single-informant approach is another limitation. Although a multi-informant approach has its merits, its response rate is a serious concern, likely resulting in a sample too small for analyzing a model as complex as ours. Given the weaknesses of the single-informant approach, we consequently applied various methods suggested by MacKenzie et al. [57], Weisberg et al. [92], and Huber and Power [41] to mitigate potential problems. Third, because power asymmetry has traditionally been a notorious feature of dyadic relationships, we cannot eliminate the confounding effect of power on some of our proposed relationships. Fourth, our variance model tested with data from a cross-sectional survey cannot offer clear-cut causality among constructs such as IOS integration, process integration, and interfirm collaboration. Finally, our definition and operationalization of IT soft skills are far from complete. For the current research, we simply extracted the essence of the literature to frame technology-management and interpersonal skills as the two primary dimensions of IT soft skills. A more comprehensive study of IT soft skills is needed.

We offer the following directions for future research. First, academics could investigate the effect of IT soft skills in other contexts because they have been so emphasized by practitioners. Second, collecting data from both sides of a supply-chain dyad would allow examination of the effect of better-matched skill levels on developing IOS. Third, understanding the relationships between the mechanisms studied in this research and other governance mechanisms, such as trust, may deserve additional research attention. Finally, exploring dimensions of the relational view other than interfirm collaboration could also be a fruitful direction for future supply-chain management research.

Appendix A. Questionnaire items

IT Soft Skills (Byrd and Turner [10]; Lee et al. [52])

Interpersonal Skills (IS)

- IS1 Our IT staff is capable of teaching others.
- IS2 Our IT staff has the ability to plan, organize, and lead projects.
- IS3 Our IT staff has the ability to plan and execute work in a collective environment.
- IS4 Our IT staff has the ability to accomplish multiple assignments.
- IS5 Our IT staff works well in cross-functional teams to address business problems.
- IS6 Our IT staff has the ability to work cooperatively in a project-team environment.
- IS7 Our IT staff has the ability to work closely with users.
- IS8 Our IT staff has the ability to write clear, concise, and effective memos, reports, and documentation.

Technology Management Skills (TSM)

- TSM1 Our IT staff is knowledgeable about the key success factors that must be present if our organization is to succeed.
- TSM2 Our IT staff is encouraged to learn new technologies and related business knowledge.
- TSM3 Our IT staff closely follows trends in current technologies.
- TSM4 Our IT staff considers IT investments as long term and consistent.
- TSM5 Our IT staff is willing to understand and employ new techniques.

IT Hard Skills (HS) (Byrd and Turner [10]; Lee et al. [52])

- HS1 Our IT staff is skilled in multiple programming languages.
- HS2 Our IT staff is skilled in distributed processing or distributed computing.
- HS3 Our IT staff is skilled in network management and maintenance.
- HS4 Our IT staff is skilled in developing web-based applications.
- HS5 Our IT staff is skilled in data warehousing, mining, or marts.

Interfirm Collaboration (CO) (Stank et al. [85]; Kim et al., [46])

- CO1 Our firm and its major supplier coordinate their efforts harmoniously.
- CO2 Our firm and its major suppliers of various departments frequently attend cross-functional meetings.
- CO3 Our firm and its major supplier meet frequently for both formal and informal discussions of important issues.
- CO4 Our firm benchmarks best practices/processes and shares results with its major suppliers.
- CO5 Our firm has supply-chain arrangements with its major supplier that operate under principles of shared risks and rewards.
- CO6 Our firm effectively shares operational information externally with its major supplier.

IOS Integration (IOSI) (Saraf et al. [75]; Rai and Tang [69])

- IOSI1 Data are entered only once for retrieval by most of our major supplier's applications.
- IOSI2 We can easily share our data with our major supplier.
- IOSI3 We have successfully integrated most of our software applications with our major supplier.
- IOSI4 Most of our software applications work seamlessly across our major suppliers.
- IOSI5 Our applications easily aggregate relevant information from our major supplier's databases (e.g., operating information, business customer performance, and cost information).
- IOSI6 Our applications have the capability to exchange real-time information with our major supplier.

Process Integration (PI) (Saraf et al. [75])

- PI1 To facilitate operations, our firm's business procedures and routines are linked with its major supplier.
- PI2 Our way of doing business is closely linked with our major supplier.
- PI3 Our firm's business procedures and routines are highly coupled with its major supplier.
- PI4 Some of our operations are closely connected with our major supplier.
- PI5 To operate efficiently, we rely on our major supplier's procedures and routines.

Supply-Chain Performance (SCP) (Klein and Rai [48]; Wang et al. [91])

Our organization has realized the following performance outcomes because of our interactions with the major supplier:

- SCP1 Improved asset management.
- SCP2 Increased productivity.
- SCP3 Lower operating costs.
- SCP4 Improved production planning.
- SCP5 Improved resource control.
- SCP6 Increased flexibility.
- SCP7 Quick placement of new product designs into production.
- SCP8 Efficient operations at different levels of output.
- SCP9 Development or modification of new product designs.
- SCP10 Production of a wide variety of product mixes simultaneously.
- SCP11 On-time responses to market demand.

Appendix B. Cross-loadings of measurement items

	IS	TSM	HS	CO	IOSI	PI	SCP
IS1	0.71	0.45	0.35	0.26	0.06	0.11	0.19
IS2	0.85	0.58	0.42	0.28	0.07	0.10	0.27
IS3	0.85	0.59	0.42	0.28	0.05	0.11	0.24
IS4	0.77	0.49	0.44	0.26	0.09	0.11	0.18
IS5	0.80	0.48	0.38	0.26	0.10	0.15	0.24
IS6	0.82	0.56	0.31	0.22	0.01	0.08	0.19
IS7	0.77	0.58	0.33	0.30	0.04	0.15	0.23
IS8	0.73	0.63	0.47	0.42	0.22	0.26	0.32
TSM1	0.62	0.70	0.43	0.43	0.28	0.19	0.32
TSM2	0.46	0.70	0.39	0.40	0.14	0.18	0.24
TSM3	0.49	0.80	0.46	0.31	0.12	0.14	0.29

Appendix B. (Continued)

	IS	TSM	HS	CO	IOSI	PI	SCP
TSM4	0.51	0.78	0.27	0.24	-0.01	0.08	0.23
TSM5	0.53	0.81	0.38	0.31	0.11	0.08	0.31
HS1	0.25	0.24	0.67	0.18	0.21	0.19	0.19
HS2	0.39	0.45	0.81	0.45	0.29	0.21	0.23
HS3	0.43	0.42	0.61	0.24	0.10	0.16	0.25
HS4	0.36	0.38	0.75	0.32	0.21	0.20	0.29
HS5	0.43	0.41	0.79	0.45	0.27	0.27	0.17
CO1	0.32	0.38	0.40	0.84	0.45	0.38	0.27
CO2	0.30	0.37	0.45	0.88	0.51	0.45	0.33
CO3	0.30	0.35	0.43	0.89	0.56	0.53	0.34
CO4	0.36	0.43	0.41	0.89	0.45	0.48	0.33
CO5	0.24	0.37	0.41	0.85	0.52	0.49	0.32
CO6	0.36	0.42	0.31	0.80	0.38	0.35	0.26
IOSI1	0.07	0.11	0.23	0.47	0.85	0.55	0.28
IOSI2	0.11	0.19	0.28	0.55	0.81	0.51	0.29
IOSI3	0.07	0.13	0.29	0.49	0.94	0.57	0.32
IOSI4	0.13	0.20	0.31	0.51	0.90	0.56	0.33
IOSI5	0.10	0.15	0.29	0.49	0.90	0.58	0.34
IOSI6	0.05	0.13	0.28	0.48	0.93	0.59	0.32
PI1	0.12	0.15	0.25	0.48	0.63	0.91	0.35
PI2	0.17	0.16	0.27	0.52	0.56	0.94	0.39
PI3	0.15	0.13	0.25	0.49	0.63	0.96	0.40
PI4	0.18	0.18	0.28	0.49	0.54	0.94	0.42
PI5	0.17	0.20	0.25	0.45	0.57	0.88	0.42
SCP1	0.21	0.29	0.21	0.24	0.26	0.39	0.77
SCP2	0.28	0.37	0.29	0.33	0.31	0.32	0.84
SCP3	0.31	0.43	0.35	0.35	0.30	0.34	0.83
SCP4	0.18	0.23	0.21	0.29	0.33	0.40	0.86
SCP5	0.23	0.25	0.26	0.32	0.29	0.39	0.87
SCP6	0.25	0.27	0.20	0.32	0.30	0.36	0.86
SCP7	0.24	0.31	0.18	0.25	0.23	0.33	0.87
SCP8	0.29	0.35	0.22	0.34	0.28	0.36	0.86
SCP9	0.28	0.36	0.29	0.33	0.38	0.37	0.85
SCP10	0.27	0.31	0.23	0.32	0.33	0.39	0.89
SCP11	0.20	0.31	0.26	0.29	0.30	0.34	0.83

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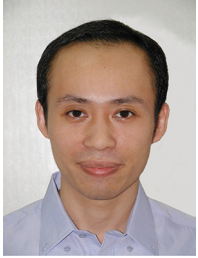
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